# Abreham Dessu

## Full Stack Software Engineer

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### PROFESSIONAL SUMMARY

Full Stack Software Engineer with **4+ years of experience** shipping complete products — from PostgreSQL schema and event-driven microservices through to the React and Next.js interfaces users touch. Scaled a Kafka-based microservices platform to **10,000+ users**, engineered payment infrastructure processing **5,000+ daily transactions at a 96% success rate**, and improved system reliability **30%** while reducing downtime **40%**. Fluent across Node.js, NestJS, TypeScript, React, Next.js, and Angular, with production deployment experience on Google Cloud Platform (GCP) and Amazon Web Services (AWS). **AWS Certified Solutions Architect – Associate.** Works fluently with AI-assisted development tooling (Claude Code, Cursor, GitHub Copilot, Codex) to accelerate delivery without sacrificing code quality or review discipline.

### CORE COMPETENCIES

Full Stack Development | RESTful API Design | GraphQL API Development | Microservices Architecture | Event-Driven Architecture | Apache Kafka | Distributed Systems | Database Design | SQL Query Optimization | Responsive Web Design | Component-Based Architecture | Server-Side Rendering (SSR) | Single Page Applications (SPA) | UI Development | Cross-Browser Compatibility | Third-Party API Integration | Webhook Handling | Authentication & Authorization | Role-Based Access Control (RBAC) | Containerization | CI/CD Pipelines | Cloud Architecture | Production Monitoring | Performance Optimization | AI-Assisted Development | System Design | Agile / Scrum

### TECHNICAL SKILLS

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|---|---|
| **Languages** | JavaScript, TypeScript, Python, Go, SQL, HTML5, CSS3, Bash |
| **Frontend** | React, Next.js, Angular, Tailwind CSS, Responsive Web Design, Component-Based Architecture, Server-Side Rendering (SSR), REST & GraphQL Client Integration, Semantic HTML, Cross-Browser Compatibility |
| **Backend** | Node.js, NestJS, Express.js, Django, Flask, REST APIs, GraphQL, Microservices, Event-Driven Architecture, Apache Kafka, Prisma ORM, Webhooks |
| **Databases & Data** | PostgreSQL, MySQL, MongoDB, DynamoDB, Meilisearch, Query Optimization, Indexing, Schema Migrations, Data Modeling |
| **Cloud & DevOps** | Google Cloud Platform (GCP), Amazon Web Services (AWS), Microsoft Azure, Docker, Kubernetes, CI/CD Pipelines, Deployment Automation |
| **Security & Auth** | Firebase Authentication, OAuth 2.0, JSON Web Tokens (JWT), Token-Based Access Control, Role-Based Access Control (RBAC) |
| **AI-Assisted Dev** | Claude Code, Cursor, GitHub Copilot, Codex, Prompt Engineering, AI Pair Programming, AI-Assisted Code Review, AI Productivity Applications, Amazon PartyRock |
| **Testing & Practices** | Unit Testing, Integration Testing, Test-Driven Development (TDD), Code Review, Production Monitoring, Incident Response, Git, GitHub, Postman, Jira, Agile / Scrum, Data Structures & Algorithms |

### PROFESSIONAL EXPERIENCE

#### Full Stack Engineer
Jun 2025 – Present

**ETM Software · Addis Ababa, Ethiopia**

- Improved system reliability **30%** by re-architecting the backend service layer and implementing production monitoring, alerting, and automated CI/CD deployment pipelines on Google Cloud Platform (GCP).
- Engineered a fault-tolerant integration keeping **5,000+ HubSpot company and contact records** synchronized with near-zero data drift, using retry logic, deduplication, and reconciliation to survive partial API failures and rate limiting.
- Eliminated manual data entry across **2,000+ hierarchical entities** by automating multi-level synchronization workflows, removing a recurring source of data corruption in business-critical records.
- Reduced feature delivery time **25%** by designing **5+ reusable service modules** in NestJS and Prisma ORM, establishing shared data-access and validation patterns across the codebase.
- Secured business-critical operations with **zero authentication-related incidents to date**, implementing Firebase Authentication with JSON Web Token (JWT) access control and role-based permissions (RBAC).
- Designed and deployed GraphQL and RESTful APIs on NestJS and PostgreSQL, collapsing multi-call data fetching into single round trips and enabling client-side workflow automation for internal teams.

#### Full Stack Developer
Sep 2023 – Jun 2025

**Ablaze Labs · Addis Ababa, Ethiopia**

- Scaled payment infrastructure to **5,000+ daily transactions at a 96% success rate** by integrating Chapa and SantimPay gateways with idempotent request handling, webhook signature verification, and automated failure recovery.
- Reduced production downtime **40%** and improved performance **20%** through query profiling, PostgreSQL index tuning, and disciplined production incident triage and root-cause analysis.
- Scaled an event-driven microservices gaming platform to **10,000+ users**, building both the NestJS/Kafka backend services and the React interfaces for user management, game management, referrals, and payments.
- Led end-to-end delivery of a national content management system for Ethiopia's **Ministry of Water and Energy**, architecting backend APIs and building the frontend content management modules across **12 interconnected subprojects**.
- Replaced manual, document-driven government reporting with automated APIs and CMS interfaces, reducing report turnaround from **3 days to under 2 hours** across multiple ministry projects.

- Increased user acquisition **35%** on a job board platform by shipping full stack employer and applicant workflows, subscription billing, and a referral-based growth engine.
- Delivered Problem-Based Learning functionality for a medical e-learning platform, building the collaborative case-session interface and supporting services used by **500+ students and instructors**.

### Junior Software Developer

Oct 2022 – Aug 2023

**Hirundo Security Solution · Addis Ababa, Ethiopia**

- Improved deployment efficiency **22%** by automating server configuration and provisioning across **10+ machines** with repeatable, scripted Bash workflows.
- Increased IT support efficiency **15%** by developing internal backend systems and administrative web applications that automated the team's highest-volume manual processes.
- Consolidated fragmented operational workflows into a unified web-based toolset, measurably reducing manual workload for business teams.
- Strengthened stability of business-critical systems through unit and integration testing, debugging, and resolution of production defects before user impact.

## PROJECTS

### Microservices Gaming Platform

*NestJS, Apache Kafka, PostgreSQL, React, Firebase Authentication, Docker*

- Event-driven backend serving **10,000+ users** across independently scalable services for user management, game management, referrals, and payments, paired with a React frontend for player onboarding, game browsing, and wallet management.

### Water & Energy CMS — Ethiopian Ministry of Water and Energy

*NestJS, PostgreSQL, React, Docker*

- Nationwide platform spanning **12 interconnected subprojects**; designed the shared backend API layer and built the content management interfaces that standardized reporting for government stakeholders.

### HubSpot Enterprise Integration Service

*NestJS, Prisma ORM, PostgreSQL, GCP*

- Bidirectional synchronization keeping **5,000+ CRM records** and **2,000+ hierarchical entities** consistent between HubSpot and internal PostgreSQL systems under real-world API rate limits and partial failures.

### Job Board & Talent Marketplace

*Next.js, NestJS, PostgreSQL, Meilisearch, Tailwind CSS*

- End-to-end employer and applicant experiences — job posting, application tracking, subscription billing, and referral-driven acquisition — with responsive server-rendered Next.js interfaces and sub-second Meilisearch faceted search.

## EDUCATION

### A2SV (African to Silicon Valley)

**Software Engineering Trainee — Data Structures and Algorithms**

Jan 2025 – Jan 2026 · Addis Ababa, Ethiopia

Selected for a highly competitive program preparing African engineers for global technology roles, focused on advanced algorithms, data structures, and system design.

### Adigrat University

**Bachelor of Science, Electrical and Computer Engineering**

Graduated Jul 2022 · Adigrat, Ethiopia

## CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate** — Amazon Web Services
- **AWS Certified Cloud Practitioner** — Amazon Web Services
- **Microsoft Certified: Azure Fundamentals (AZ-900)** — Microsoft
- **AWS Scholars Program Project Badge: Analyze Data using AI with PartyRock** — Udacity, Mar 2026
- **AWS Scholars Program Project Badge: Build Your First AI Productivity App** — Udacity, Mar 2026
- **Professional Foundations** — ALX Africa
- **AI Career Essentials** — ALX Africa